Lecture 27

Solitons and topological charge

Objectives. *Lecture 26 classified maps into an order-parameter space. A field theory also assigns an energy to each configuration. We separate the boundary conditions that define a topological sector from the field equations that select stationary configurations. The kink supplies a complete energy-bound calculation. The gauge vortex shows how a phase winding becomes a quantized magnetic flux.*

By the end of this lecture you will be able to:

- Identify a sector from its stated continuous vacuum boundary data.
- Derive the kink bound and its saturating profile with the signed charge.
- Prove Derrick's obstruction for the stated linear-target scalar energy, including $d = 2$.
- Derive vortex flux quantization under the stated asymptotic conditions.

Unit 27A Kinks, boundary sectors and scalar scaling

27.1 The scalar kink and the Bogomolny bound

Definition 27.1. A *static soliton* is a spatially localized, finite-energy stationary solution of the field equations.

Here static means time independent. Localization concerns the energy density: the field itself may approach different nonzero vacua at the two ends.

Take one real field on the line and constants $\mu, v > 0$.

Physical input. We choose the relativistic scalar model with Lagrangian density $\mathcal{L} = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - V(\phi)$, in units with light speed one. Its energy density is $\frac{1}{2}\phi_t^2 + \frac{1}{2}\phi_x^2 + V(\phi)$. The potential is the double well $V(\phi) = \frac{1}{2}\mu^2(v^2 - \phi^2)^2$.

Recall the field Euler–Lagrange equation: differentiating this Lagrangian gives

$$\phi_{tt} - \phi_{xx} + V'(\phi) = 0.$$

A static field therefore satisfies $\phi'' = V'(\phi)$. We choose a smooth function W whose derivative squares to twice this potential. This auxiliary function is called a *superpotential*. Define W and the static energy by

$$W(\phi) = \mu\left(v^2\phi - \frac{\phi^3}{3}\right), \qquad V(\phi) = \frac{1}{2}\big(W'(\phi)\big)^2, \qquad E = \int_{\mathbb{R}} \left(\frac{1}{2}(\phi')^2 + V(\phi)\right) \mathrm{d}x.$$

Here W' differentiates with respect to the field value, while ϕ' differentiates with respect to x.

Definition 27.2. The zero set $\mathcal{V} = V^{-1}(0)$ is the *vacuum manifold* when it is a manifold.

The vacua of this double well are $-v$ and $+v$. The potential and the kink's localized energy density are shown in the next figure.

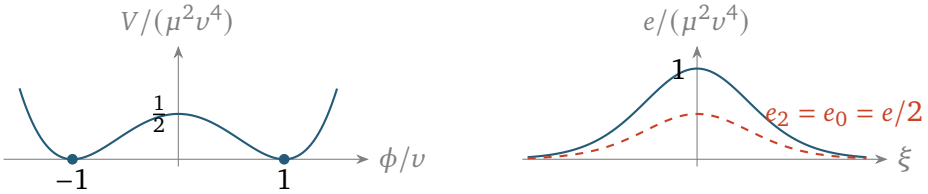


Figure 27.1: The double well vanishes at $\phi/v = \pm 1$. The kink has energy density $e/(\mu^2 v^4) = \operatorname{sech}^4 \xi$, where $\xi = \mu v(x - x_0)$, with equal gradient and potential halves.

Assume vacuum limits ϕ_- and ϕ_+ at the two ends:

$$\lim_{x \to -\infty} \phi(x) = \phi_-, \qquad \lim_{x \to +\infty} \phi(x) = \phi_+, \qquad \phi_-, \phi_+ \in \{-v, +v\}.$$

The figure shows the increasing kink profile centred at zero.

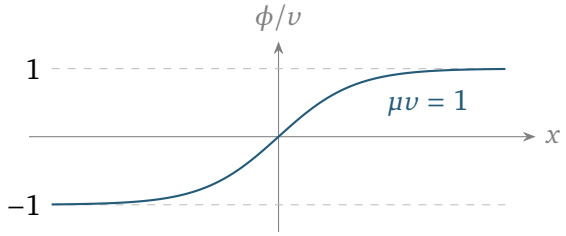


Figure 27.2: The exact kink $\phi/v = \tanh(\mu v x)$ joins $-v$ to $+v$. The increasing profile has charge +1.

Definition 27.3. A *soliton sector* fixes the boundary homotopy data, which here are the ordered vacuum endpoints.

Definition 27.4. A *topological charge* is an invariant used to label such a sector.

Definition 27.5. A *Bogomolny bound* is an energy bound fixed by boundary topological data.

Definition 27.6. The first-order equation $\phi' = sW'(\phi)$, for $s \in \{+1, -1\}$, is the *Bogomolny equation* for this model.

For this model, the difference of the endpoint values of W gives a Bogomolny bound.

Proposition 27.7 (The kink bound · *proved*). *Every smooth finite-energy configuration with these limits satisfies $E \geq |W(\phi_+) - W(\phi_-)|$. In the sector $\phi_- = -v$, $\phi_+ = v$, equality is attained by*

$$\phi_{\text{kink}}(x) = v \tanh\big(\mu v(x - x_0)\big), \qquad x_0 \in \mathbb{R}, \qquad E_{\text{kink}} = \frac{4\mu v^3}{3}.$$

Its charge $Q = (\phi_+ - \phi_-)/(2v)$ is +1. The decreasing antikink has charge -1 and the same energy.

Proof. For either sign $s \in \{+1, -1\}$, completing the square gives

$$E = \frac{1}{2}\int_{\mathbb{R}} (\phi' - sW'(\phi))^2 \mathrm{d}x + s\big(W(\phi_+) - W(\phi_-)\big).$$

The last term follows from the chain rule on a finite interval and then the stated limits. Choose s so its product with this difference is nonnegative. The square is nonnegative, proving the bound. Equality requires $\phi' = sW'(\phi)$, the Bogomolny equation.

For the increasing sector, this equation is $\phi' = \mu(v^2 - \phi^2)$. Separating variables on $-v < \phi < v$ gives $\operatorname{artanh}(\phi/v) = \mu v(x - x_0)$, hence the displayed profile. Its derivative is $\mu v^2 \operatorname{sech}^2(\mu v(x - x_0)) = W'(\phi)$. It has the required endpoint limits, so it saturates the bound. Direct endpoint evaluation gives $W(v) - W(-v) = 4\mu v^3/3$. Differentiating the first-order equation gives $\phi'' = W''W' = V'(\phi)$, the static field equation. Thus this saturating configuration is a solution, not only a trial field. Changing its sign gives the antikink and reverses the ordered vacuum pair and charge. ∎

Equal endpoint vacua give charge zero. The bound then permits the vacuum configuration. The square completion proves a global energy minimum in each kink sector. It does not imply that every configuration in that sector is stationary.

The static equation also gives a first integral, because

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(\phi')^2 - V(\phi)\right) = \phi'(\phi'' - V'(\phi)) = 0.$$

Finite energy gives a sequence tending to infinity along which ϕ' tends to zero. The vacuum limit makes $V(\phi)$ tend to zero. Thus the constant in this first integral is zero. Along the kink we obtain $\frac{1}{2}(\phi')^2 = V(\phi)$. The gradient and potential contributions to its energy are equal.

For this double well, no nonconstant finite-energy static solution has equal endpoint vacua. Indeed, a nonconstant field with equal limits attains an interior maximum or minimum different from that common limit. At this turning point the first integral forces $V = 0$, so the field is at one of the vacua with derivative zero. ODE uniqueness for the smooth static equation then forces the constant vacuum solution. This conclusion uses the field equation and its first integral, rather than charge zero alone.

The kink also gives a moving localized solution. For a speed $\beta \in (-1, 1)$, define

$$\phi_\beta(t, x) = \phi_{\mathrm{kink}}\big(\gamma(x - \beta t)\big), \qquad \gamma = (1 - \beta^2)^{-1/2}.$$

The wave operator applied to this field gives $\gamma^2(\beta^2 - 1)\phi''_{\mathrm{kink}} = -\phi''_{\mathrm{kink}}$, so the static equation proves that the moving field solves the same evolution equation. Write E_2 and E_0 for the kink's rest-frame gradient and potential energies. Changing variables to $\gamma(x - \beta t)$ in the energy gives

$$E_\beta = \gamma(1 + \beta^2)E_2 + \frac{E_0}{\gamma} = \gamma(E_2 + E_0), \qquad E_2 = E_0.$$

The minimum energy in the ordered kink boundary class is positive while the continuous vacuum endpoints remain fixed. The translating energy density and the relativistic energy formula give the kink a particle interpretation, with rest mass equal to its rest energy in these units.

27.2 Vacuum boundary data and sectors

Recall that a classical field configuration assigns a value to each spatial point. For a scalar field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, take a smooth potential $V \geq 0$. Its vacuum set is again $\mathcal{V} = V^{-1}(0)$. A constant field in $\mathcal{V}$ has zero potential and gradient energy.

In one dimension, assume limits ϕ_- and ϕ_+ in $\mathcal{V}$ at $-\infty$ and $+\infty$. Call their path components $\mathcal{V}_-$ and $\mathcal{V}_+$. In dimensions two and three, assume a continuous angular limit $\phi_\infty : S^{d-1} \to \mathcal{V}$, with convergence uniform in direction.

For an evolving field, require that its limiting boundary map be jointly continuous in angle and time. In one dimension require each endpoint limit to be continuous in time. The potential assigns positive energy density to values outside $\mathcal{V}$. This motivates vacuum limits, which we impose as boundary assumptions. Finite energy alone does not imply them. The vacuum manifold $\mathcal{V}$ is the order-parameter space $\mathcal{M}$ of Lecture 26 in these models. The sphere at infinity surrounds the entire localized region. A linking sphere from Lecture 26 carries the same class if it can be deformed to this sphere through a region on which the normalized field stays defined. In one dimension the sphere S^0 consists of the two distinguished endpoints. A homotopy of its maps moves each endpoint within its path component. The figure compares these boundary maps in the kink, vortex and hedgehog examples.

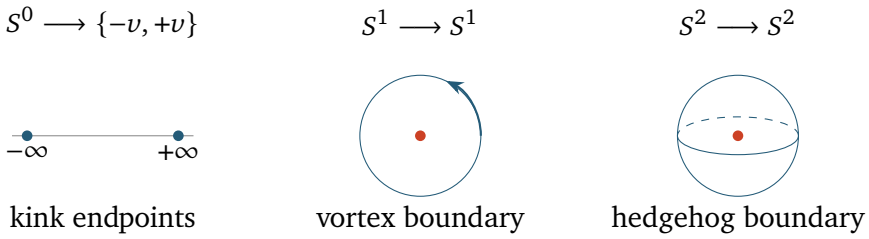


Figure 27.3: The boundary spheres have dimensions zero, one and two. Their maps take values in the vacuum set or target sphere of the corresponding model.

The figure preserves the order of the one-dimensional endpoints.

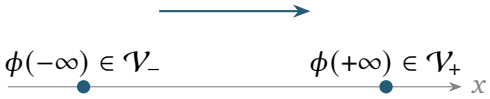


Figure 27.4: The left and right vacuum components form an ordered pair. Interchanging the two ends can reverse a kink's charge.

Proposition 27.8 (Boundary sectors · *proved*)**.** *Under these boundary conditions, the one-dimensional datum is the ordered pair of path components of $\mathcal{V}$ containing ϕ_- and ϕ_+. In dimensions two and three it is the free homotopy class of ϕ_∞. An evolution with jointly continuous vacuum boundary data preserves these data.*

Proof. At either end in one dimension, the time-dependent limiting value is a path in $\mathcal{V}$. It cannot leave its path component. The two ends remain distinguished, giving an ordered pair. For $d = 2$ or $d = 3$, the jointly continuous boundary map is exactly a homotopy of maps $S^{d-1} \to \mathcal{V}$. Its free homotopy class is therefore fixed. Lecture 26 expresses these classes using conjugacy or the fundamental-group action when needed. ∎

An integer charge exists in the kink and vortex examples, but the general sector datum need not be an integer. It constrains allowed deformations. Energy and the field equations determine whether a soliton exists.

A texture has different data: its value at infinity is fixed, and the degree belongs to the map on the compactified whole plane, as in Lecture 26. Its constant boundary map alone does not determine that degree.

27.3 Derrick's scaling obstruction

For a smooth linear-target field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, use

$$E = E_2 + E_0, \qquad E_2 = \frac{1}{2}\int_{\mathbb{R}^d} |\nabla\phi|^2 \mathrm{d}^d x, \qquad E_0 = \int_{\mathbb{R}^d} V(\phi)\mathrm{d}^d x, \qquad V \geq 0.$$

The subscript counts derivatives in the energy term. A spatial dilation is $\phi_\lambda(x) = \phi(\lambda x)$ for $\lambda > 0$. The chain rule contributes λ^2 to the gradient term. The change of variables contributes λ^{-d} to both terms. Thus

$$E(\lambda) = \lambda^{2-d}E_2 + \lambda^{-d}E_0, \qquad E'(1) = (2-d)E_2 - dE_0.$$

The figure shows this expression with $E_2 = E_0 = 1$.

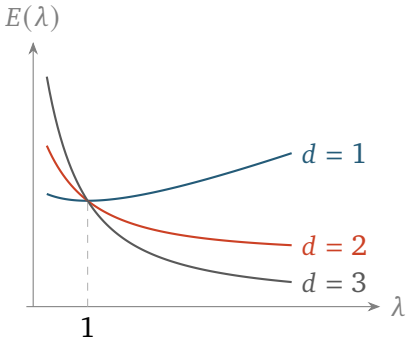


Figure 27.5: Dilation energy for the stated linear-target scalar model. Only the one-dimensional curve has a stationary point for positive E_2 and E_0.

Proposition 27.9 (Derrick's theorem · *proved*). *Let $V\colon \mathbb{R}^N \to [0,\infty)$ be smooth. Every smooth finite-energy static solution $\Delta\phi = \nabla V(\phi)$ of this model on $\mathbb{R}^d$, for $d \geq 2$, is a constant vacuum.*

Proof. **Step 1: the stress identity.** The field equation gives stationarity against compactly supported variations. A dilation changes the field arbitrarily far away, so this fact does not yet justify $E'(1) = 0$. Its infinitesimal variation is $x^j\partial_j\phi$, where x^j are the spatial coordinates. We cut off this variation at large radius. The stress tensor writes the resulting integration by parts as a divergence. Use the Euclidean metric to raise indices and let $e = |\nabla\phi|^2/2 + V(\phi)$. The stress tensor and its divergence are

$$T^i{}_j = \partial^i\phi\cdot\partial_j\phi - \delta^i_j e, \qquad \partial_i T^i{}_j = (\Delta\phi - \nabla V(\phi))\cdot\partial_j\phi = 0.$$

To see the cancellation, differentiate the first term by the product rule. Its second-derivative term $\partial^i\phi\cdot\partial_i\partial_j\phi$ is the derivative of $|\nabla\phi|^2/2$ and cancels that part of $\partial_j e$. The remaining potential derivative is $\nabla V\cdot\partial_j\phi$.

Step 2: the cutoff limit. Fix one smooth bump χ from Lecture 21, equal to one on the unit ball and zero outside the radius-two ball. Put $\chi_R(x) = \chi(x/R)$. The chain rule gives

$|\nabla\chi_R| \le C/R$ with $C = \sup|\nabla\chi|$. Integrate the divergence of $T^i{}_j x^j \chi_R$ and use compact support to obtain

$$\int_{\mathbb{R}^d} \chi_R T^i{}_i \mathrm{d}^d x = -\int_{\mathbb{R}^d} T^i{}_j x^j \partial_i \chi_R \mathrm{d}^d x.$$

The absolute value of the right side is bounded by a constant times $\int_{R<|x|<2R}(|\nabla\phi|^2 + V)\mathrm{d}^d x$, which tends to zero by finite energy. The trace is integrable, so dominated convergence on the left gives $(2-d)E_2 - dE_0 = 0$. For $d > 2$, nonnegativity forces $E_2 = E_0 = 0$, hence a constant vacuum.

Step 3: dimension two. For $d = 2$ the gradient term has zero scaling weight. The identity gives $E_0 = 0$ and leaves E_2 undetermined. We use the field equation to control this remaining term. Continuity and nonnegativity imply $V(\phi(x)) = 0$ everywhere. Each such field value minimizes V, so $\nabla V(\phi(x)) = 0$. The field equation becomes $\Delta\phi = 0$. Let u be any component of a derivative $\partial_i\phi$. It is harmonic and square-integrable. Its spherical mean about any point p is independent of radius: its derivative is the normalized outward flux of ∇u, zero by Stokes and $\Delta u = 0$. The mean tends to $u(p)$ at radius zero, so averaging the sphere means radially gives the ball mean. Cauchy–Schwarz therefore yields

$$|u(p)| = \left|\frac{1}{|B_R|}\int_{B_R(p)} u \mathrm{d}^d x\right| \le |B_R|^{-1/2}\|u\|_{L^2(\mathbb{R}^d)} \longrightarrow 0.$$

Thus all derivatives vanish and ϕ is a constant vacuum in this case too. ∎

A constrained sphere-valued sigma model is outside the linear-target hypothesis: its field equation is not $\Delta\phi = \nabla V$ for a potential on all of $\mathbb{R}^N$. Lecture 26's degree-one plane texture has this constrained target. Gauge fields and higher-derivative energies also have different scaling powers. They need not satisfy this obstruction. Our dilation is $\phi(\lambda x)$.[1] In one dimension the same stationary scaling identity reads $E_2 = E_0$, as for the kink.

For a smooth unit-vector field $\mathbf{n}\colon \mathbb{R}^2 \to S^2$ extending smoothly to the compactified plane, consider the finite Dirichlet energy

$$E_\sigma = \frac{1}{2}\int_{\mathbb{R}^2} (|\partial_x \mathbf{n}|^2 + |\partial_y \mathbf{n}|^2)\mathrm{d}x\mathrm{d}y.$$

This energy is invariant under spatial dilation in dimension two. Since $\mathbf{n}\cdot\partial_i\mathbf{n} = 0$, the cross product with $\mathbf{n}$ preserves the length of $\partial_i\mathbf{n}$. For $s \in \{+1, -1\}$ we therefore have

$$|\partial_x \mathbf{n} - s\mathbf{n}\times\partial_y\mathbf{n}|^2 = |\partial_x\mathbf{n}|^2 + |\partial_y\mathbf{n}|^2 + 2s\mathbf{n}\cdot(\partial_x\mathbf{n}\times\partial_y\mathbf{n}).$$

Lecture 26 identifies the integral of the last density with $4\pi Q$. Integration gives

$$E_\sigma = \frac{1}{2}\int_{\mathbb{R}^2} |\partial_x\mathbf{n} - s\mathbf{n}\times\partial_y\mathbf{n}|^2 \mathrm{d}x\mathrm{d}y - 4\pi s Q \ge 4\pi|Q|$$

when $s = -\operatorname{sgn}(Q)$ for nonzero charge. For zero charge, either sign gives the bound zero. For Lecture 26's degree-one texture of scale $a > 0$, differentiation gives

$$|\partial_x\mathbf{n}|^2 = |\partial_y\mathbf{n}|^2 = \frac{4a^2}{(a^2+x^2+y^2)^2}, \qquad \partial_x\mathbf{n} = -\mathbf{n}\times\partial_y\mathbf{n}.$$

Thus its energy is 4π for every scale a. This family attains the lower bound, so it is stationary under smooth compactly supported variations within the unit sphere. Its size is not fixed by

[1] Manton–Sutcliffe, equations (4.17)–(4.19), pp. 84–85, use this dilation. Inverting the scale parameter changes the derivative sign but not the obstruction.

this energy. For Lecture 26's three-dimensional hedgehog, the same Dirichlet model instead has $|\nabla \mathbf{n}|^2 = 2/r^2$. The energy in $r_0 < r < R$ is $4\pi(R - r_0)$, which diverges as R grows.

Self-check. For the scalar kink model, list the possible values of $Q = (\phi_+ - \phi_-)/(2v)$ for vacuum endpoints. Count the ordered endpoint sectors and decide whether the charge distinguishes all of them.

Answer. The four ordered pairs are $(-v, -v)$, $(-v, +v)$, $(+v, -v)$ and $(+v, +v)$. Their charges are respectively zero, +1, −1 and zero. The two equal-endpoint pairs are different sectors because a continuous vacuum-valued endpoint cannot move between the disconnected vacua. Thus the charge is an invariant, but is not a complete label of the sector.

Self-check. State in one sentence what a static soliton is and why we introduced its energy. Explain why a nonzero charge is insufficient to identify one.

Answer. A static soliton is a localized finite-energy stationary field, and its energy distinguishes stationary profiles and sector minima among fields with the same boundary topology. A charge labels allowed deformations, whereas the field equation must still be satisfied.

Unit 27B Gauge vortices and monopole charge

27.4 The abelian Higgs vortex and its flux

Definition 27.10. A *Higgs field* here is a charged complex scalar $\phi\colon \mathbb{R}^2 \to \mathbb{C}$.

The symbol ϕ denotes the field, and θ denotes the polar angle.

Physical input. We use the static abelian Higgs energy below as a field model. In its Ginzburg–Landau interpretation, the complex field describes an electron-pair condensate in a superconductor (Manton–Sutcliffe, Chapter 1 and §7.1). The pair charge is $q = -2e$, where $e > 0$ is the elementary charge. For fields constant along a straight vortex line, the plane is a transverse section and E is energy per unit length.

Let $q \neq 0$ be its signed charge and put $a = q/\hbar$. Use a smooth real potential A_{em} on the plane and Lecture 21's derivative $D = \mathrm{d} - \mathrm{i}aA_{\mathrm{em}}$. Write $A_{\mathrm{em}} = A_x \mathrm{d}x + A_y \mathrm{d}y$ and $B = \partial_x A_y - \partial_y A_x$. The symbol $v > 0$ again denotes the vacuum magnitude. Here $\lambda > 0$ is the scalar coupling, distinct from the dilation parameter in the preceding section. The static abelian Higgs energy is

$$E = \int_{\mathbb{R}^2} \left[\frac{1}{2} B^2 + \frac{1}{2}\left(|D_x\phi|^2 + |D_y\phi|^2\right) + \frac{\lambda}{8}\left(|\phi|^2 - v^2\right)^2 \right] \mathrm{d}x\mathrm{d}y.$$

Its vacuum field values have modulus v, a circle. Write $r = \sqrt{x^2 + y^2}$. Without a gauge field, the winding tail $\phi = ve^{\mathrm{i}n\theta}$ outside a core has gradient energy

$$E_{\mathrm{tail}} = \frac{1}{2} \int_{r_0}^{R} \int_0^{2\pi} \frac{v^2 n^2}{r^2} r \mathrm{d}\theta \mathrm{d}r = \pi v^2 n^2 \log(R/r_0), \qquad 0 < r_0 < R.$$

For nonzero winding this diverges as $R \to \infty$. The gauge potential permits cancellation of the angular phase derivative in $D\phi$. This cancellation makes finite planar energy possible.

The gauge model also changes the scaling argument. Write E_B, E_D and E_V for its magnetic, covariant-gradient and potential terms. For a dilation factor $\kappa > 0$, put $\phi_\kappa(x) = \phi(\kappa x)$ and

let $(A_{\mathrm{em}})_\kappa$ be the pullback of A_{em} under $x \mapsto \kappa x$. The potential components gain one factor of κ, and the magnetic field gains two. In dimension two the energy therefore scales as

$$E(\kappa) = \kappa^2 E_B + E_D + \kappa^{-2} E_V, \qquad E'(1) = 2(E_B - E_V).$$

The two weighted terms can balance, in contrast to the scalar obstruction. The figure shows the exact curve for $E_B = E_D = E_V = 1$.

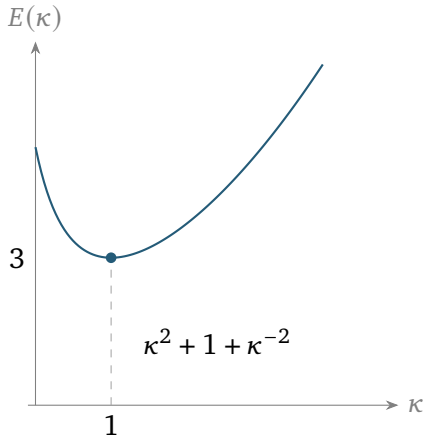


Figure 27.6: The magnetic and potential energies have opposite scaling weights. Their sum has a minimum at $\kappa = 1$ for these equal coefficients.

For $r = R$, write $\phi_R(\theta) = \phi(R\cos\theta, R\sin\theta)$. The symbol D_θ means the covariant derivative along the angular parameter, so $D_\theta \phi_R = \partial_\theta \phi_R - \mathrm{i}a(A_{\mathrm{em}})_\theta \phi_R$. It includes the radius factor relative to a unit tangent derivative.

The winding tail suggests a vanishing angular covariant derivative. We impose the precise integrated condition in the theorem as a boundary assumption.

Theorem 27.11 (Vortex flux · *sketched; flux proved*). *Assume B is integrable, $|\phi_R| \to v$ uniformly, and $\int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \to 0$ as $R \to \infty$. Let n be the winding of $\phi_R/|\phi_R|$ for large R. The total flux is*

$$\Phi = \int_{\mathbb{R}^2} B \mathrm{d}x\mathrm{d}y = \frac{2\pi n}{a} = \frac{2\pi\hbar n}{q}.$$

At the critical coupling *$\lambda = a^2$, finite-energy fields in this sector obey $E \geq \pi v^2 |n|$.*

Flux proof. The amplitude is bounded below by $v/2$ outside a sufficiently large disk. Thus the nonzero fields on any two large circles are homotopic through the intervening annulus, and their winding n agrees. On each circle, the imaginary part of the logarithmic derivative gives

$$a \oint_{r=R} A_{\mathrm{em}} = 2\pi n - \int_0^{2\pi} \mathrm{Im}\left(\frac{D_\theta \phi_R}{\phi_R} \right) \mathrm{d}\theta.$$

The last integral tends to zero by the amplitude lower bound and the assumed angular derivative integral. Stokes on the disk gives $\oint A_{\mathrm{em}} = \int_{r \leq R} B \mathrm{d}x\mathrm{d}y$. Integrability of B gives the total flux in the limit, proving the formula. ∎

These asymptotic conditions are part of the sector. Finite energy alone was not used to infer pointwise limits. The magnetic flux now has the same integer origin as Lecture 3's winding and Lecture 22's quantization. Lecture 21 gives the limiting holonomy $e^{\mathrm{i}a\Phi} = e^{2\pi \mathrm{i} n} = 1$. Thus

the limiting Aharonov–Bohm phase of the condensate is trivial. For $q = -2e$, the magnitude of a single flux quantum is

$$\frac{2\pi\hbar}{2e} = \frac{h_{\mathrm{P}}}{2e}, \qquad h_{\mathrm{P}} = 2\pi\hbar.$$

A nowhere-zero field on a disk gives a disk extension of its normalized boundary map. Lecture 3 then gives winding zero. Consequently nonzero winding forces at least one zero of the Higgs field in the plane. Gauge maps in this discussion are smooth on the whole plane. Their restriction to a circle contracts through the disk to the value at its centre. Its winding is therefore zero, so these gauge changes preserve n.

For a sign $s \in \{+1, -1\}$, we obtain the bound by writing squares plus a boundary term, as for the kink. The square $|(D_x + \mathrm{i}sD_y)\phi|^2$ produces a mixed derivative term. The commutator converts that term into a curl and a multiple of $B|\phi|^2$. At critical coupling the magnetic and potential terms form a second square whose mixed term cancels this multiple. Only a flux term remains after integration.

Let $j_i = \mathrm{Im}(\bar{\phi} D_i \phi)$, for $i \in \{x, y\}$, be the components of a real boundary-current one-form, distinct from the physical current of Lecture 15. Set $\mathrm{curl}\, j = \partial_x j_y - \partial_y j_x$. On a radius-$R$ circle its circulation satisfies

$$\left| \oint_{r=R} j \right| \leq \sup_{\theta} |\phi_R| \int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \longrightarrow 0.$$

The sector assumptions therefore remove this boundary term. Using $[D_x, D_y]\phi = -\mathrm{i}aB\phi$ gives $\mathrm{curl}\, j = 2\,\mathrm{Im}(\overline{D_x\phi} D_y \phi) - aB|\phi|^2$. For either sign $s \in \{+1, -1\}$, the resulting square completion at $\lambda = a^2$ is

$$E = \frac{1}{2} \int_{\mathbb{R}^2} \left[|(D_x + \mathrm{i}sD_y)\phi|^2 + \left(B + \frac{sa}{2}(|\phi|^2 - v^2) \right)^2 \right] \mathrm{d}x\mathrm{d}y + \frac{sav^2}{2}\Phi.$$

Obtain this identity first on disks using Stokes for the curl term. The boundary-current circulation tends to zero by the estimate above. Finite energy and integrability of the magnetic field give the limit. Choosing $s = \mathrm{sgn}(n)$ for $n \neq 0$ gives the bound. The first-order equations that set both squares to zero are

$$(D_x + \mathrm{i}sD_y)\phi = 0, \qquad B = -\frac{sa}{2}(|\phi|^2 - v^2).$$

Their solutions saturate the bound. Their existence requires an analytic input. Away from the zeros of ϕ, these first-order equations reduce to one nonlinear elliptic equation for $\log|\phi|^2$. The prescribed zeros supply its point-source conditions. Solving that boundary problem constructs vortices. The analytic existence step is cited from Manton–Sutcliffe, §7.8, pp. 199–200. Their normalization is $a = v = 1$, with critical $\lambda = 1$ and bound $\pi|n|$.[2]

The figure shows the smooth winding-one trial field calculated in the example, for $a > 0$. Its radial arrows show the phase as a direction in the internal complex plane. The separately labelled tangent arrow gives the circulation of A_{em}.

[2] In general their connection potential is aA_{em}. Its curvature is aB, so potential rescaling alone does not equate their unit-coupling energy with the general-a energy printed here.

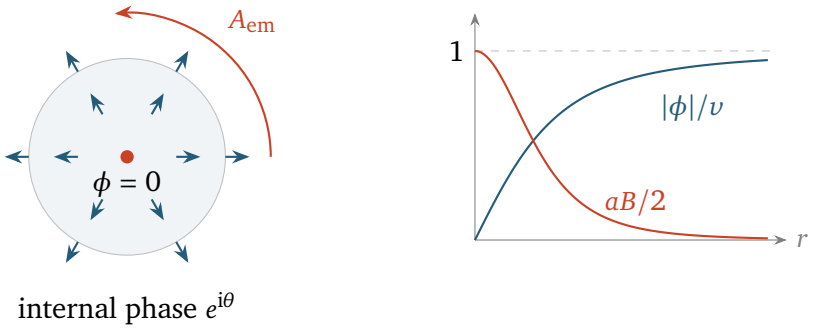


Figure 27.7: The trial field has internal phase $e^{i\theta}$ and positive potential circulation for $a > 0$. The exact radial profiles approach vacuum amplitude and zero magnetic field, respectively; this trial field is not a solution of the first-order equations.

Example 27.12 (From asymptotic phase to flux). Assume $\phi_R \to v e^{in\theta}$ with its first angular derivative, and $D_\theta \phi_R \to 0$ uniformly. The limiting derivative equation then gives $(A_{\mathrm{em}})_\theta \to n/a$ in this gauge. With B integrable, the flux proof needs only the integrated angular condition, and gives $\Phi = 2\pi n/a$. A negative signed charge reverses the magnetic flux for the same positive winding.

For a concrete smooth configuration of winding one, use

$$\phi(x,y) = \frac{v(x+iy)}{\sqrt{1+x^2+y^2}}, \qquad A_{\mathrm{em}} = \frac{-y\,dx + x\,dy}{a(1+x^2+y^2)}, \qquad B = \frac{2}{a(1+x^2+y^2)^2}.$$

It has a Higgs zero at the origin and no singular potential there. On a radius-R circle, $\phi_R = vRe^{i\theta}/\sqrt{1+R^2}$ and $(A_{\mathrm{em}})_\theta = R^2/[a(1+R^2)]$. Thus $D_\theta \phi_R = i\phi_R/(1+R^2)$ tends uniformly to zero. The flux is

$$\Phi = \frac{4\pi}{a}\int_0^\infty \frac{r\,dr}{(1+r^2)^2} = \frac{2\pi}{a}.$$

Direct differentiation of the displayed fields gives

$$D_x\phi = \frac{v}{(1+r^2)^{3/2}}, \qquad D_y\phi = \frac{iv}{(1+r^2)^{3/2}}.$$

Thus the sum of the squared norms of these two derivatives is $2v^2/(1+r^2)^3$. Substitution of $t = 1+r^2$ makes the radial integrals of $r/(1+r^2)^p$ equal to $1/[2(p-1)]$, for $p \in \{2,3,4\}$. Each energy density term is integrable. The total energy is

$$E = \pi\left[\frac{2}{3a^2} + \frac{v^2}{2} + \frac{\lambda v^4}{8}\right].$$

At critical coupling we can compare this trial with the bound in a fixed theory by varying its width $\ell > 0$. Replace its fields by

$$\phi_\ell = \frac{v(x+iy)}{\sqrt{\ell^2+r^2}}, \qquad (A_{\mathrm{em}})_\ell = \frac{-y\,dx + x\,dy}{a(\ell^2+r^2)}.$$

The same radial integrals give

$$E(\ell) = \pi\left[\frac{2}{3a^2\ell^2} + \frac{v^2}{2} + \frac{a^2v^4\ell^2}{8}\right].$$

The first and last terms have the form $A/\ell^2 + C\ell^2 \geq 2\sqrt{AC}$. They are equal at the minimizing width, so

$$\ell^2 = \frac{4}{\sqrt{3}\,a^2v^2}, \qquad E_{\text{trial,min}} = \pi v^2\left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right) > \pi v^2.$$

The field demonstrates the finite-energy flux sector but does not attain its Bogomolny bound. It is a trial configuration, not an asserted solution of the vortex field equations.

27.5 The monopole charge *(optional)*

Lecture 28's zero-mode count does not depend on this section.

Remark 27.13 (The Yang–Mills–Higgs monopole)**.** In the SU(2) Yang–Mills–Higgs model, an adjoint Higgs field is a vector in the Lie algebra that transforms by conjugation. Its fixed-norm vacuum directions form $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. With a continuous nonzero Higgs limit at infinity, the normalized field defines a map from the outward spatial sphere to this vacuum sphere. Its degree is the topological monopole charge $Q \in \mathbb{Z}$, as in Lecture 26. Orient the Lie algebra by Lecture 23's basis T^1, T^2, T^3. The hedgehog with coefficients x^a/r in that basis restricts to the identity direction map between outward spheres, so it has $Q = +1$. The remaining U(1) connection acts on charged line bundles at infinity. Lecture 22 then requires $q\Phi_{\mathrm{m}} = 2\pi\hbar n$ for each such charge convention. Relating that integer to the degree also requires specifying the chosen charged representation.

Physical input. The SU(2) Yang–Mills–Higgs equations admit smooth finite-energy monopoles (Manton–Sutcliffe, §8.2). Their Higgs amplitude vanishes at the core while the normalized field approaches the hedgehog at infinity. The residual abelian field approaches a Dirac monopole field at large distance. These are cited properties of those solutions.

Lecture 22's abelian description has a singular central point. The Yang–Mills–Higgs solution instead has smooth full fields at its core. The following figure compares only their direction maps on a surrounding sphere.

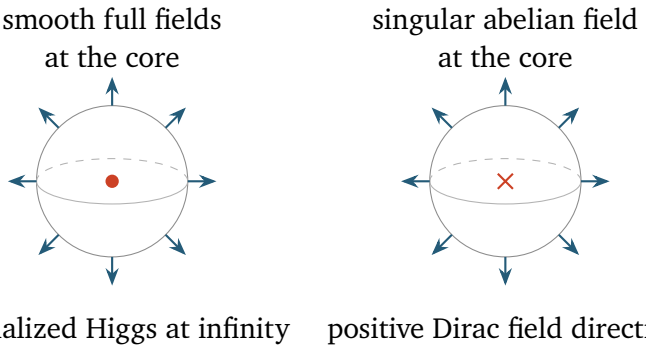


Figure 27.8: The normalized asymptotic Higgs map is the outward identity direction map of degree +1. A positive-flux Dirac field also has outward directions, but its abelian field is singular at the origin.

The degree describes the monopole boundary topology. The analytic monopole construction and its mass bound are treated in Manton–Sutcliffe, §8.2.

Self-check. A vortex has winding $n = -3$ and signed charge $q = -2e$ with $e > 0$. Determine its signed flux and critical-coupling energy bound. Then consider the printed width-one trial field in a different model, with $a = -2$ and $v = 3$ at critical coupling. Test the magnetic first-order equation at the origin and decide whether this trial attains its bound.

Answer. Substitution of both negative signs gives $\Phi = 3\pi\hbar/e = 3h_{\mathrm{P}}/(2e)$, and the bound is $E \geq 3\pi v^2$. The trial has winding +1, so its saturating sign is $s = +1$. At the origin its field vanishes and its magnetic field is $B(0) = 2/a = -1$. The first-order equation instead requires $B(0) = sav^2/2 = -9$. Thus this trial does not attain the bound, although it satisfies the sector assumptions.

Summary. Continuous vacuum boundary data define sectors that an evolution with jointly continuous vacuum boundary data cannot change. Finite energy alone does not supply those limits. The scalar kink saturates $E \geq |\Delta W|$ and has signed charge +1. Derrick's theorem excludes nonconstant smooth finite-energy static solutions of the stated nonnegative-potential linear-target scalar model for $d \geq 2$. The gauge vortex lies outside that model and has flux $2\pi\hbar n/q$ under the stated asymptotic conditions. Its critical-coupling bound follows by square completion. Existence uses the cited analytic input. Lecture 28 uses a quantized sphere flux to count chiral quantum zero modes.

Reading. Manton–Sutcliffe, Chapters 1 and 6; §4.2, pp. 84–85; §5.1–5.2, pp. 110–113; §7.3, pp. 164–165; §7.8, pp. 197–200; §8.2.